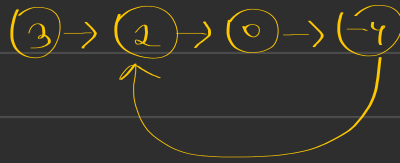


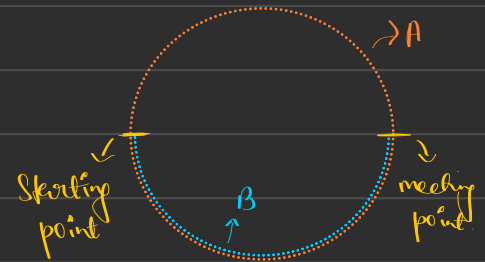
Detect loop :-



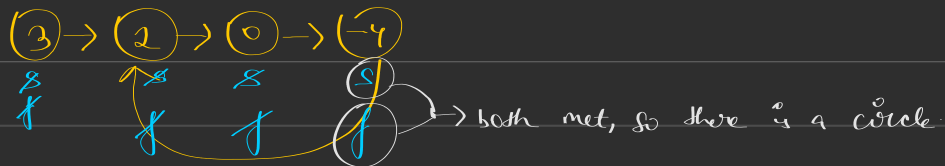
Suppose there is a ground, two people A and B are running.

Speed of A is 2 times the speed of B.

If both of them starts the race from the same point, A and B will meet at the point $\text{perimeter}/2$, so you can say that the ground is round in shape. If it was a st. road, they would've never met.



Applying the same logic,



$T.C \rightarrow O(n)$
 $S.C \rightarrow O(1)$